

SRM University – AP, Andhra Pradesh
Neeru Konda, Mangalagiri Mandal
Guntur District, Mangalagiri, Andhra Pradesh 522240

Name of the Course: Explainable Artificial Intelligence

Course Code	AML 572	Course Category	Core Elective (CW)	L-T-P-C	3	0	0	3
Pre-Requisite Course(s)		Co-Requisite Course(s)		Progressive Course(s)				
Course Offering Department	CSE	Professional / Licensing Standards						

Course Objectives / Course Learning Rationales (CLRs)

Objective 1. Learn the foundations of Explainable AI and understand various interpretability frameworks including intrinsic and post-hoc methods.

Objective 2. To introduce the algorithmic foundations of feature attribution, counterfactual reasoning, and concept-based explanation.

Objective 3. To discuss issues that make explanation a hard task — faithfulness, disagreement, robustness, and adversarial vulnerability.

Objective 4. To discuss well-known applications of XAI in medicine, finance, law, NLP, and the frontier of mechanistic interpretability.

Course Outcomes / Course Learning Outcomes (CLOs)

	At the end of the course the learner will be able to	Bloom's Level	Expected Proficiency Percentage	Expected Attainment Percentage
Outcome 1	Recall the fundamental concepts of interpretability, explainability, and transparency.	1	70%	68%
Outcome 2	Demonstrate algorithms for feature attribution, saliency analysis, and counterfactual generation.	2	70%	65%
Outcome 3	Develop systems for concept-based reasoning and global explanation of black-box models.	3	70%	60%
Outcome 4	Implement and evaluate mechanistic interpretability methods on transformer-based language models.	4	70%	65%

Course Articulation Matrix (CLO) to Program Learning Outcomes (PLO)

CLOs	Eng ineer ing Know ledge	Desi gn / Dev elop ment of Solu tion s	Con duct Inve stig ation s of C omp lex Pro ble ms	Con duct Inve stig ation s of C omp lex Pro ble ms	Mod ern Tool Usa ge	The Engi neer and Soci ety	Env iron ment and Sust aina bilit y	Ethi cs	Indi vidu al and Tea m Wo rk	Co mm unic ation	Life -lon g Le arn ing	PS O 1	PS O 2	PS O 3
Outco me 1	2	3	3		2							3	2	
Outco me 2	2	3	3		2							2	2	
Outco me 3	2	3	2		2							2	2	
Outco me 4	3	3	3		2							2	3	
Course Avera ge	2	3	3		2							2	2	

Course Utilization Plan - Theory

Unit No.	Unit Name	Required Contact Hours	CLOs Addressed	References Used
Unit 1		9		
	Introduction and Overview: What is XAI; Why explanation is difficult: faithfulness vs plausibility; Regulatory drivers (GDPR, EU AI Act).	2	1	1,2
	Definitions: Interpretability, Explainability, Transparency, Simulatability, Decomposability (Lipton's taxonomy).	2	1	1,2
	Doshi-Velez & Kim evaluation framework: Application-grounded, Human-grounded, Functionally-grounded.	1	1	1,3
	Intrinsic interpretable models: Linear, GAM, EBM, decision trees, risk scores; Rudin's argument for high-stakes domains.	2	1,2	1,4
	Taxonomy of XAI: Local vs Global, Intrinsic vs Post-hoc, Model-specific vs Model-agnostic.	2	1,2	1,2
Unit 2		8		
	LIME: Local Interpretable Model-Agnostic Explanations; perturbation, kernel weighting, sparse linear surrogate.	2	1,2	5,6
	Shapley values from cooperative game theory; the four axioms; KernelSHAP, TreeSHAP, DeepSHAP.	2	1,2	5,6
	Anchors and rule-based local explanations; Influence functions (Koh & Liang).	1	1,2	5,6
	Gradient-based attribution: Vanilla saliency, SmoothGrad, Integrated Gradients (Sundararajan), DeepLIFT, LRP.	2	1,2	5,7
	Grad-CAM and class-discriminative visualization for CNNs; Grad-CAM++, Score-CAM, HiResCAM.	1	1,2	5,7
Unit 3		7		
	Concept-based methods: TCAV (Kim et al.), ACE, CRAFT, Network Dissection.	2	2,3	5,8
	Concept Bottleneck Models; Prototype networks (ProtoPNet); This-looks-like-that explanations.	1	2,3	5,8

Unit No.	Unit Name	Required Contact Hours	CLOs Addressed	References Used
	Counterfactual explanations: Wachter et al. formulation; DiCE, FACE, MACE, GeCo.	2	2,3	5,8
	Algorithmic recourse and the causal turn (Karimi et al.); Contrastive explanations.	1	2,3	5,8
	Global methods: PDP, ICE, ALE; Permutation importance; Surrogate models; Functional ANOVA.	1	2,3	5
Unit 4		9		
	Attention Is Not Explanation debate; Jain & Wallace vs Wiegrefe & Pinter.	1	2,3	9
	Attention Rollout, Attention Flow, Generic Transformer Attribution (Chefer et al.).	2	3,4	9
	Probing classifiers and control tasks (Hewitt & Liang); Logit Lens and Tuned Lens.	1	3,4	9,10
	Mechanistic Interpretability: Distill Circuits, Olah et al.; Mathematical Framework for Transformer Circuits.	2	3,4	10,11
	Induction heads (Anthropic 2022); Indirect Object Identification circuit; Causal tracing and ROME knowledge editing.	2	3,4	10,11
	Sparse Autoencoders (SAEs), Monosemanticity, superposition hypothesis; Automated Circuit Discovery (ACDC).	1	3,4	10,11
Unit 5		12		
	Evaluation metrics: Faithfulness, Plausibility, Robustness; Deletion/Insertion AUC; ROAR.	2	3,4	5,12
	Sanity checks (Adebayo et al.); Infidelity & Sensitivity (Yeh et al.); ERASER benchmark; Quantus toolkit.	2	3,4	5,12
	Adversarial XAI: Fooling LIME and SHAP (Slack et al.); Fairwashing; Disagreement problem (Krishna et al.).	2	3,4	5,12
	Human-Centered XAI: Miller's social-science insights; Trust calibration; Bansal et al. on AI-team performance.	2	4	5,12
	XAI in Medicine: Caruana's pneumonia case, DeGrave shortcut learning; Concept-based clinical translation.	1	4	5,12
	XAI in Finance & Law: ECOA adverse action, EU AI Act, COMPAS debate, fair-prediction impossibility (Chouldechova, Kleinberg).	1	4	5,12
	XAI for LLMs: Faithful CoT, hallucination explanation, citation grounding, sycophancy.	1	4	10,11
	Research Frontiers: SAEs at scale, ZKML + Verifiable Explanations, Decentralized AI, Causal XAI.	1	4	10,11,12
	Total Contact Hours	45		

Course Utilization Plan – Suggested In-Class Projects

Exp No.	Experiment Name	CLOs Addressed	References Used
1	Train an interpretable baseline (Logistic Regression + EBM) on a tabular dataset; report coefficients.	1	1,4
2	Compare EBM, XGBoost and Random Forest on accuracy vs interpretability tradeoffs.	1,2	1,4
3	Implement KernelSHAP from scratch and validate against the shap library on 5 test samples.	2,3	5,6
4	Apply TreeSHAP and LIME to an XGBoost model; document where they disagree.	2,3	5,6
5	Implement Anchors for high-precision rule explanations on the Adult Income dataset.	2,3	5
6	Compute vanilla saliency and SmoothGrad heatmaps on a pretrained ResNet-50.	2,3	5,7
7	Implement Integrated Gradients with multiple baselines; visualize differences.	2,3	5,7
8	Apply Grad-CAM and Grad-CAM++ to ImageNet samples; compare class-discriminative behavior.	2,3	5,7
9	Run Adebayo's model-randomization and data-randomization sanity checks on 4 saliency methods.	3,4	5,12
10	Build a 'stripes' concept activation vector; run TCAV on a zebra-vs-spaniel classifier.	3,4	5,8
11	Generate counterfactual explanations using DiCE for denied loan applicants; audit for fairness.	3,4	5,8
12	Compare PDP, ICE and ALE on a dataset with correlated features; explain divergence.	3	5
13	Use TransformerLens to reproduce induction-head behavior on GPT-2 Small.	4	9,10,11
14	Browse a public Sparse Autoencoder (Neuronpedia); ablate one feature and measure output change.	4	10,11
15	Build a verifiable explanation prototype: prove SHAP value of a small NN with EZKL.	4	10,11,12

Recommended Resources

1. Molnar, C. (2025). Interpretable Machine Learning: A Guide for Making Black Box Models Explainable (3rd ed). Online: christophm.github.io/interpretable-ml-book/
2. Lipton, Z. C. (2016). The Mythos of Model Interpretability. ICML WHI Workshop / Communications of the ACM 2018.
3. Doshi-Velez, F., Kim, B. (2017). Towards a Rigorous Science of Interpretable Machine Learning.
4. Rudin, C. (2019). Stop Explaining Black Box Machine Learning Models for High Stakes Decisions and Use Interpretable Models Instead. Nature Machine Intelligence.
5. Ribeiro, M., Singh, S., Guestrin, C. (2016). "Why Should I Trust You?": Explaining the Predictions of Any Classifier. ACM KDD.
6. Lundberg, S., Lee, S. (2017). A Unified Approach to Interpreting Model Predictions (SHAP). NeurIPS.
7. Sundararajan, M., Taly, A., Yan, Q. (2017). Axiomatic Attribution for Deep Networks (Integrated Gradients). ICML.
8. Selvaraju, R. et al. (2017). Grad-CAM: Visual Explanations from Deep Networks. ICCV.
9. Wiegrefe, S., Pinter, Y. (2019). Attention Is Not Not Explanation. EMNLP.
10. Elhage, N. et al. (2021). A Mathematical Framework for Transformer Circuits. Anthropic.
11. Olsson, C. et al. (2022). In-Context Learning and Induction Heads. Anthropic.
12. Adebayo, J. et al. (2018). Sanity Checks for Saliency Maps. NeurIPS.

Other Resources

1. <https://christophm.github.io/interpretable-ml-book/> (Molnar — Interpretable ML, free)
2. <https://transformer-circuits.pub/> (Anthropic Transformer Circuits Thread)
3. <https://distill.pub/2020/circuits/zoom-in/> (Distill Circuits Thread)
4. <https://neelnanda.io/mechanistic-interpretability> (Neel Nanda's Mech-Interp guide)
5. Kaggle Micro-Course: Machine Learning Explainability (<https://www.kaggle.com/learn/machine-learning-explainability>)
6. <https://github.com/understandable-machine-intelligence-lab/Quantus> (Quantus evaluation toolkit)

Learning Assessment

Bloom's Level of Cognitive Task	Continuous Learning Assessments (50%)								End Semester Exam (50%)	
	CLA-1 (10%)		Mid-1 (15%)		CLA-2 (10%)		Mid-2 (15%)			
	Th	Prac	Th	Prac	Th	Prac	Th	Prac	Th	Prac
Level 1 Remember / Understand	70%		65%		40%		50%		40%	
Level 2 Apply / Analyse	30%		35%		40%		50%		60%	
Level 3 Evaluate / Create					20%					
Total	100%		100%		100%		100%		100%	